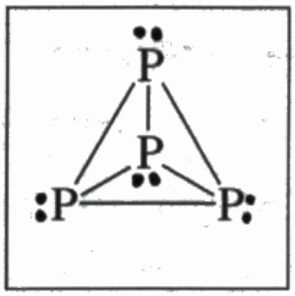
Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 3

Part A



Part B

i.) As the reaction progresses, more solid P_4O_{10} will form. Solids will create less collisions and they are more orderly than the gas O_2. Since there will be more order, the amount of disorder, spontaneousness, will decrease therefore be negative.

ii.) No, the student is incorrect. ΔG is favorable when it is a negative value. All factors including both enthalpy and entropy contribute to the favorability. Since the ΔH is negative and the ΔS is also negative, the reaction will be favorable at low temperatures. ($\Delta G = \Delta H - T\Delta S$)

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Question 3 is continued on the next page.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 3

Part C

i.) $\boxed{q = 0.159 \text{ kJ}}$

$q = (100)(4.18)(22.38 - 22)$

$4.18 \text{ J} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = 0.00418$

ii.) $\boxed{\Delta H^\circ_{rxn} = 11.2}$

$\frac{0.159 \text{ kJ}}{1 \text{ mol } P_4O_{10}} \cdot \frac{1 \text{ mol}}{283.95} \cdot \frac{4 \text{ mol}}{1 \text{ mol}}$

$\frac{0.159 \text{ kJ}}{1 \text{ mol}} \cdot \frac{283 \text{ g}}{1 \text{ mol}} \cdot \frac{1 \text{ mol}}{4 \text{ mol}} = 11.25$

Part D

Equal to the mass of P_4O_{10}, regardless of being the limiting reactant does not affect the change in temprature of the trials.

Question 3 is continued on the next page.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 3

Part E

$\frac{1}{4}P_{4(s)} + \frac{6}{4}Cl_{2(g)} \rightarrow PCl_3$ $\times\frac{1}{4}$

$PCl_3 + \frac{1}{2}Cl_2 \rightarrow PCl_5$

$\boxed{\Delta H_f^\circ = -375 \text{ kJ/mol}}$

$(-1148 \times \frac{1}{4}) + (-88) = -375$ kJ/mol

Part F

$2PCl_3 + Cl_2 \rightleftharpoons 4PCl_5$

i.) $K_p = \frac{4}{(6)(2)}$

$\boxed{K_p = 0.33}$

ii.) $\boxed{\text{increase}}$, as temperature increases, pressure increases. A higher temp will increase the pressure cause they are directly porportional.

Continue to Question 4.